



Year 10 Mathematics
Inequalities
62 Marks 60 Minutes

Topics Assessed:

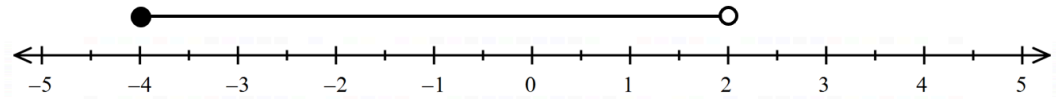
- Rearranging Inequalities
- Solving Inequalities
- Graphing on Number Lines
- Graphing in the Cartesian Plane

Name: _____

Score: / 62

1)

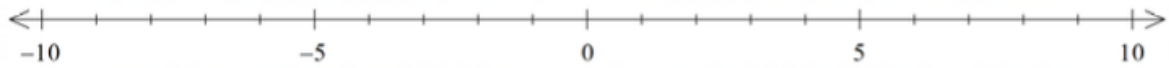
a) State the inequality represented by the number line below: [1]



b) Solve the following inequalities: [5]

$13 - 2x \leq 1$	$5(2t - 3) > 3(5 - 2t) + 18$
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2) Solve the inequality $3x + 19 > 7$ and graph the solution on the number line below. [4]



3) Solve the following [3]

$$\frac{4(2-5x)}{3} \leq 8$$

- 4) Draw a graph of $2x - 3 < 7$ on the number line below [2]



- 5) Annie, Beth and Carly go shopping. Annie spends m dollars. Beth spends twice as much as Annie. Carly spends 5 dollars more than Annie. The total amount of money spent, in dollars, is more than \$60. [6]
- a) Write down, in terms of m , an inequality to show this information.

Each girl spends a whole number of dollars.

- b) Work out the least each girl could have spent.

- 6) Lee is y years old. Toby is 8 years younger than Lee. The sum of their ages is less than 41 [5]

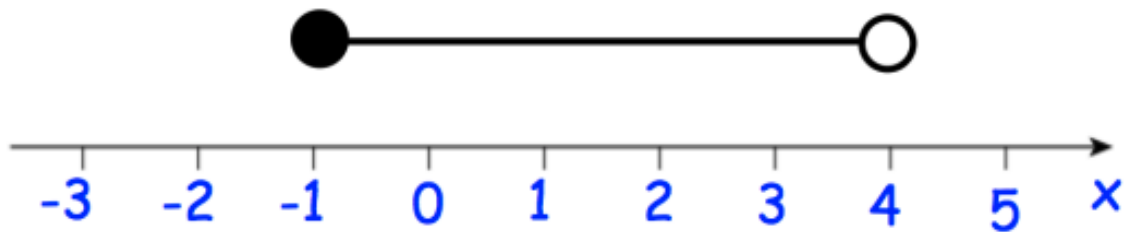
a) Write down in terms of y , an inequality to show this information.

b) Work out the oldest age that Lee can be. Give your answer as a whole number of years.

7) Given x is an integer, write all the solutions to [3]

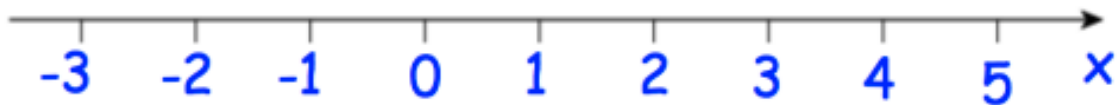
$$3 < 2x + 1 < 13$$

8) Write down the inequality shown by the diagram. [2]

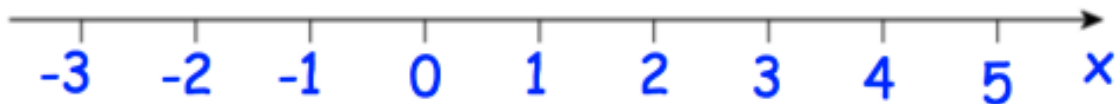


9) Represent the following on the number lines below. [4]

a) $x > 2$



b) $x \leq 4$



10) $-4 \leq n < 1$ n is an integer. [4]

a) Write down all the possible values of n

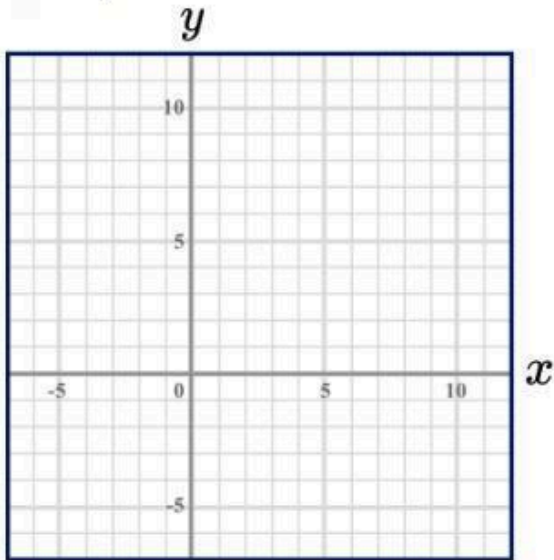
b) Solve the inequality $4x + 11 < 27$

11) a) Solve the inequality $9x + 4 < 5x - 14$ [5]

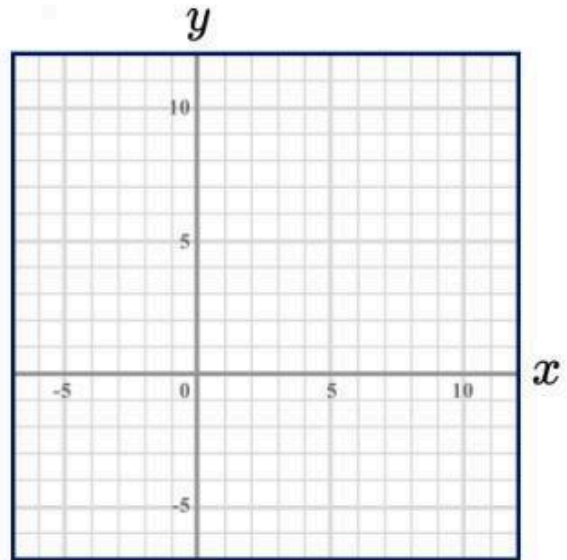
b) Given y is an integer. Write down all the solutions of the inequality $-8 \leq 2y < 0$.

12) Shade the regions that satisfy the following inequalities: [12]

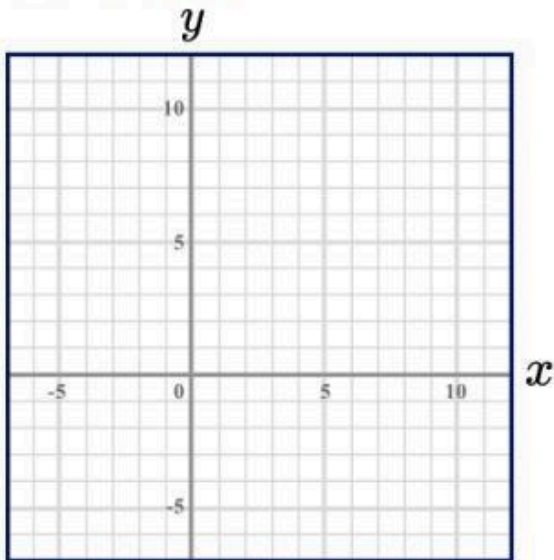
$$2 < y < 4$$



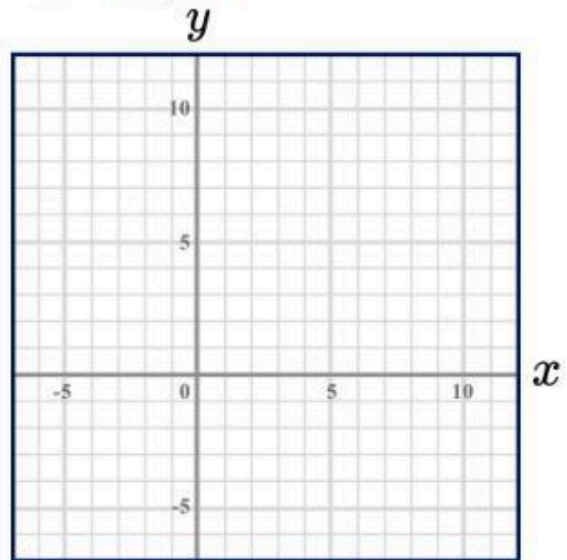
$$-2 \leq x \leq 6$$



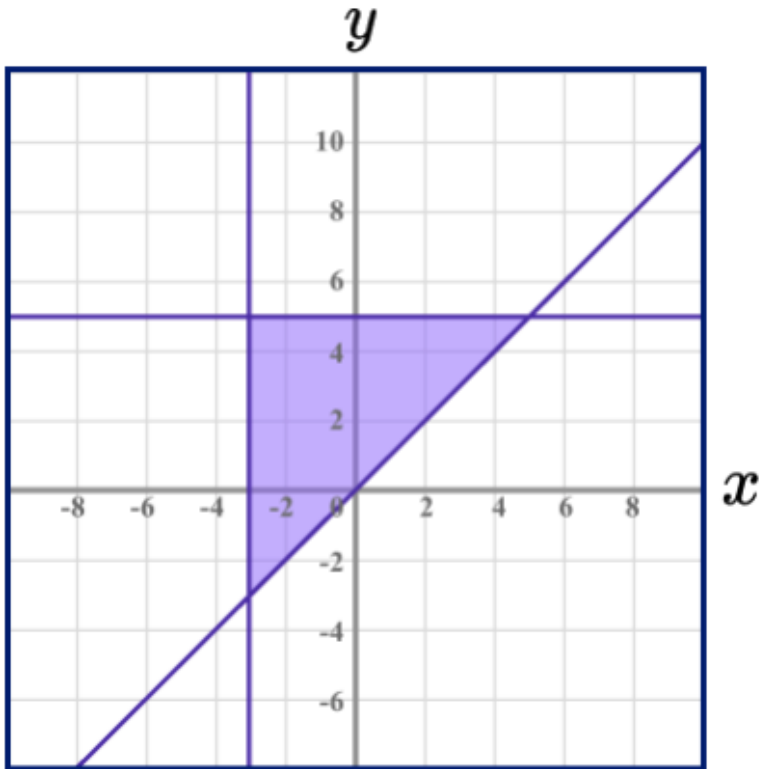
$$-1 < x \leq 6$$



$$-2 \leq y < 5$$

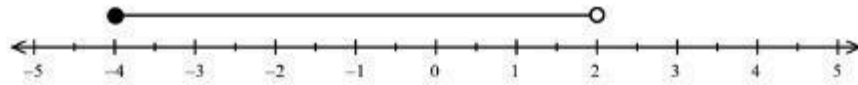


- 13) Below is a graph showing a shaded region A . Find the three inequalities which satisfy the shaded region A . [6]



1)

a) State the inequality represented by the number line below:



$$-4 \leq x < 2$$

b) Solve the following inequalities:

$$13 - 2x \leq 1$$

$$-2x \leq -12 \quad \checkmark$$

$$x \geq 6 \quad \checkmark$$

$$5(2t - 3) > 3(5 - 2t) + 18$$

$$10t - 15 > 15 - 6t + 18 \quad \checkmark$$

$$16t > 48 \quad \checkmark$$

$$t > \frac{48}{16}$$

$$t > 3 \quad \checkmark$$

2) Solve the inequality $3x + 19 > 7$ and graph the solution on the number line below. [4]

$$3x > -12 \quad \checkmark$$

$$x > -4 \quad \checkmark$$



3) Solve the following [3]

$$\frac{4(2-5x)}{3} \leq 8$$

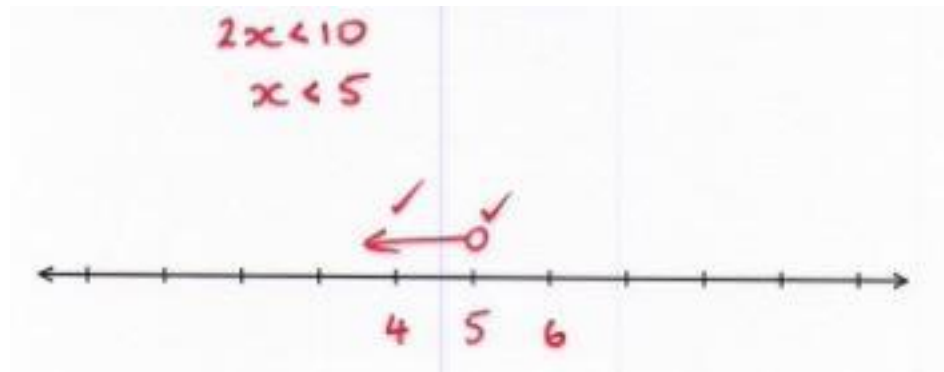
$$4(2-5x) \leq 24$$

$$2-5x \leq 6 \quad \checkmark$$

$$-5x \leq 4 \quad \checkmark$$

$$x \geq -\frac{4}{5} \quad \checkmark$$

4) Draw a graph of $2x - 3 < 7$ on the number line below [2]



5) Annie, Beth and Carly go shopping. Annie spends m dollars. Beth spends twice as much as Annie. Carly spends 5 dollars more than Annie. The total amount of money spent, in dollars, is more than \$60. [6]

a) Write down, in terms of m , an inequality to show this information.

$$4m + 5 > 60$$

Each girl spends a whole number of dollars.

b) Work out the least each girl could have spent.

$$4m > 55$$

$$m > 13.75$$

6) Lee is y years old. The sum of their ages is 1

Annie £14

Beth £28

Carly £19

than Lee. The

- a) Write down in terms of y , an inequality to show this information.

$$2y - 8 < 41$$

- b) Work out the oldest age that Lee can be. Give your answer as a whole number of years.

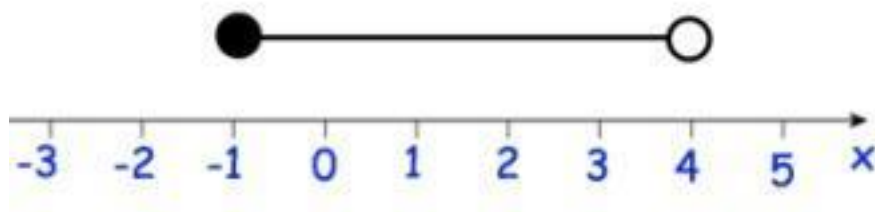
$$\begin{aligned} 2y &< 49 \\ y &< 24.5 \end{aligned}$$

- 7) Given x is an integer, write all the solutions to [3]

24

$$3 < 2x + 1 < 13$$

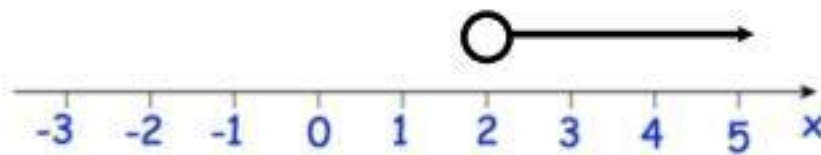
8) Write down the inequality shown by the diagram. [2]



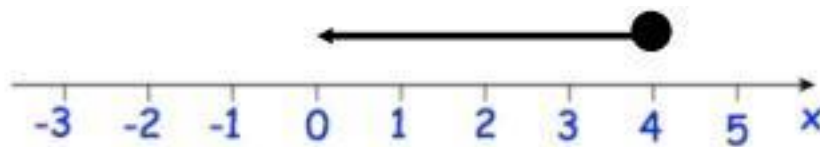
$$-1 \leq x < 4$$

9) Represent the following on the number lines below. [4]

a) $x > 2$



b) $x \leq 4$



10) $-4 \leq n < 1$ n is an integer. [4]

(a) Write down all the possible values of n .

$$-4 \ -3 \ -2 \ -1 \ 0$$

(2)

(b) Solve the inequality

$$4x + 11 < 27$$

$$4x < 16$$

11) a) Solve the inequality $9x + 4 < 5x - 14$ [5]

$$\begin{aligned} 4x + 4 &< -14 \\ 4x &< -18 \\ x &< -\frac{18}{4} \end{aligned}$$

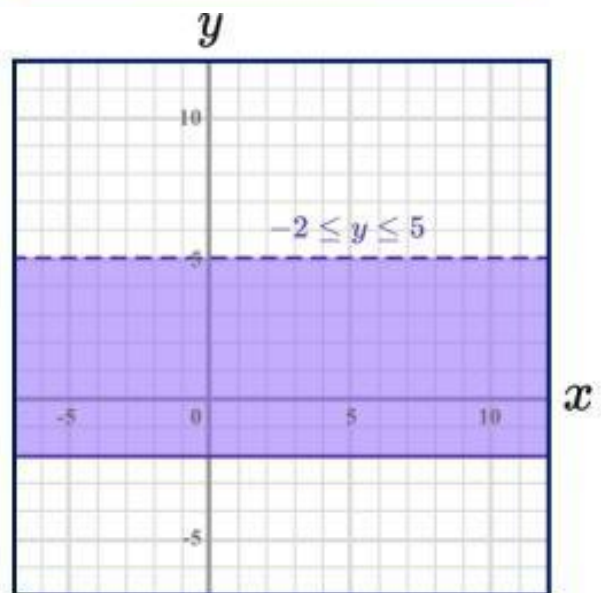
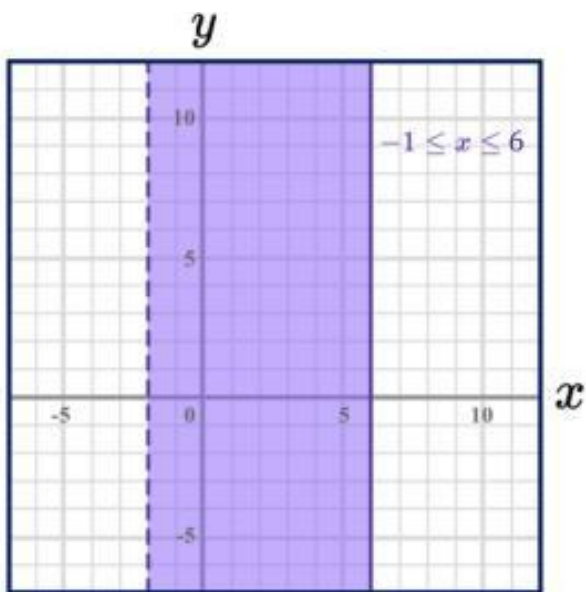
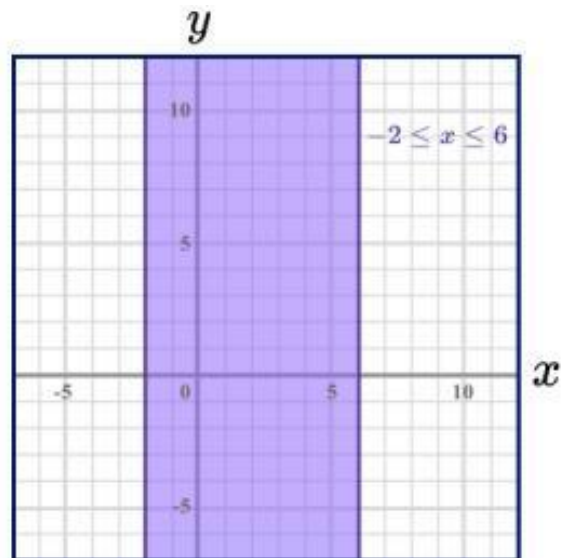
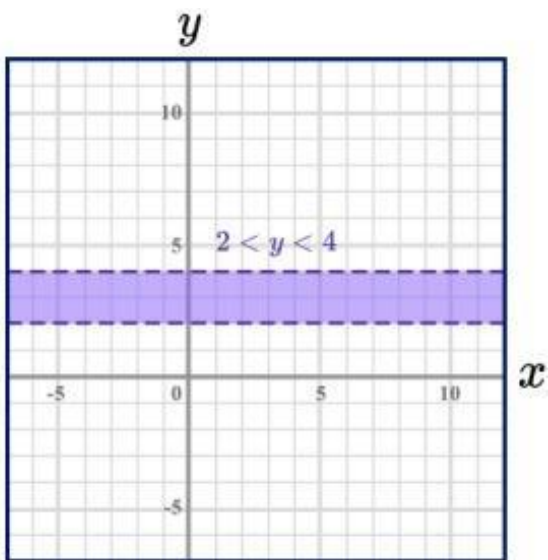
$$\begin{aligned} x &< -\frac{9}{2} \\ x &< -4.5^{(2)} \end{aligned}$$

b) Given y is an integer. Write down all the solutions of the inequality $-8 \leq 2y < 0$.

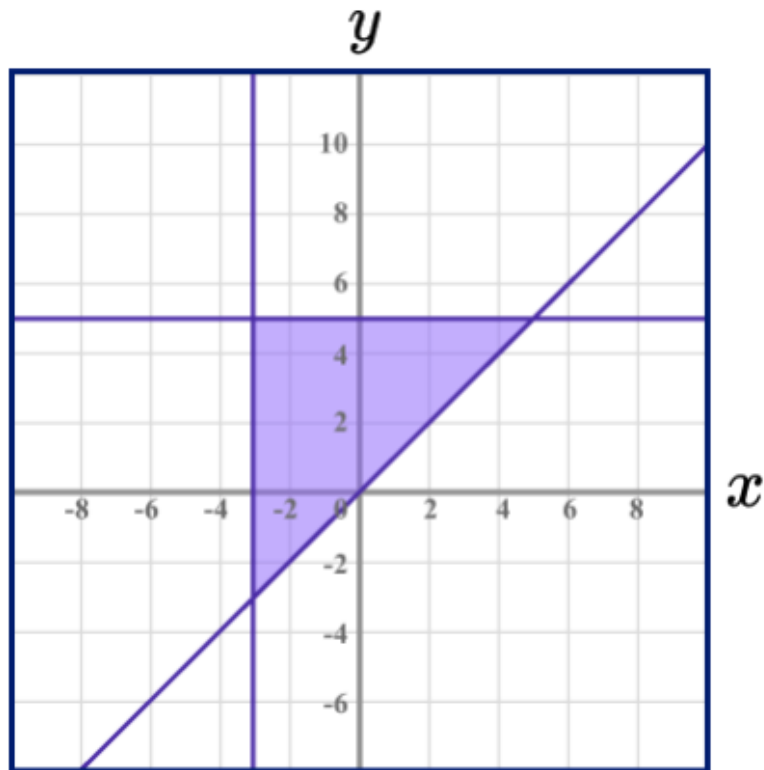
$$-4 \leq y < 0$$

$$-4 \quad -3 \quad -2 \quad -1$$

12) Shade the regions that satisfy the following inequalities: [12]



- 13) Below is a graph showing a shaded region A . Find the three inequalities which satisfy the shaded region A . [6]



$$x > -2, y \geq x \text{ and } y < 5$$